

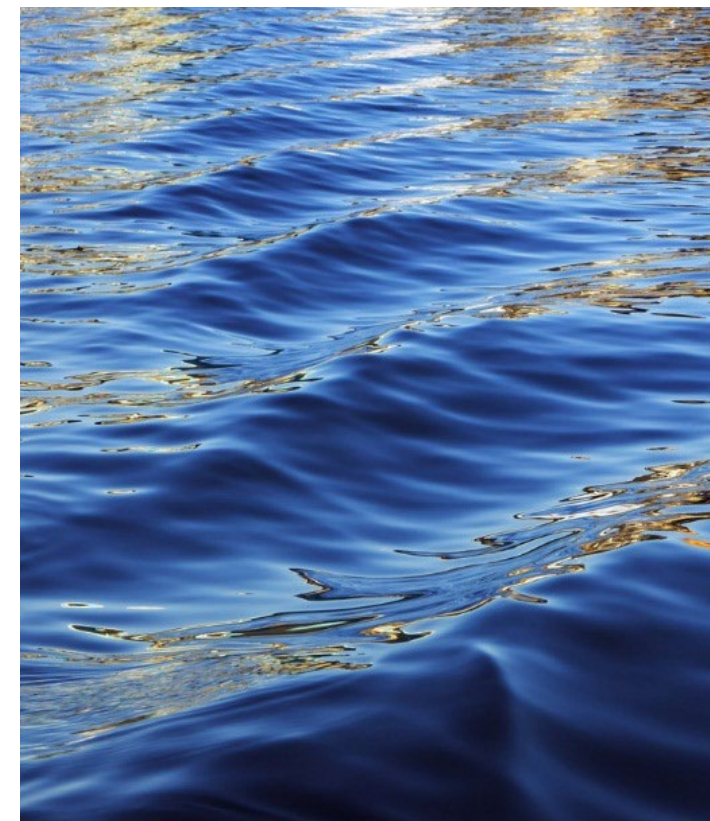


Risks Strategy



Miguel A. López-
Carmona
Alvaro Paricio-
García

@copyright



Co-funded by the
European Union



Universidad
de Alcalá

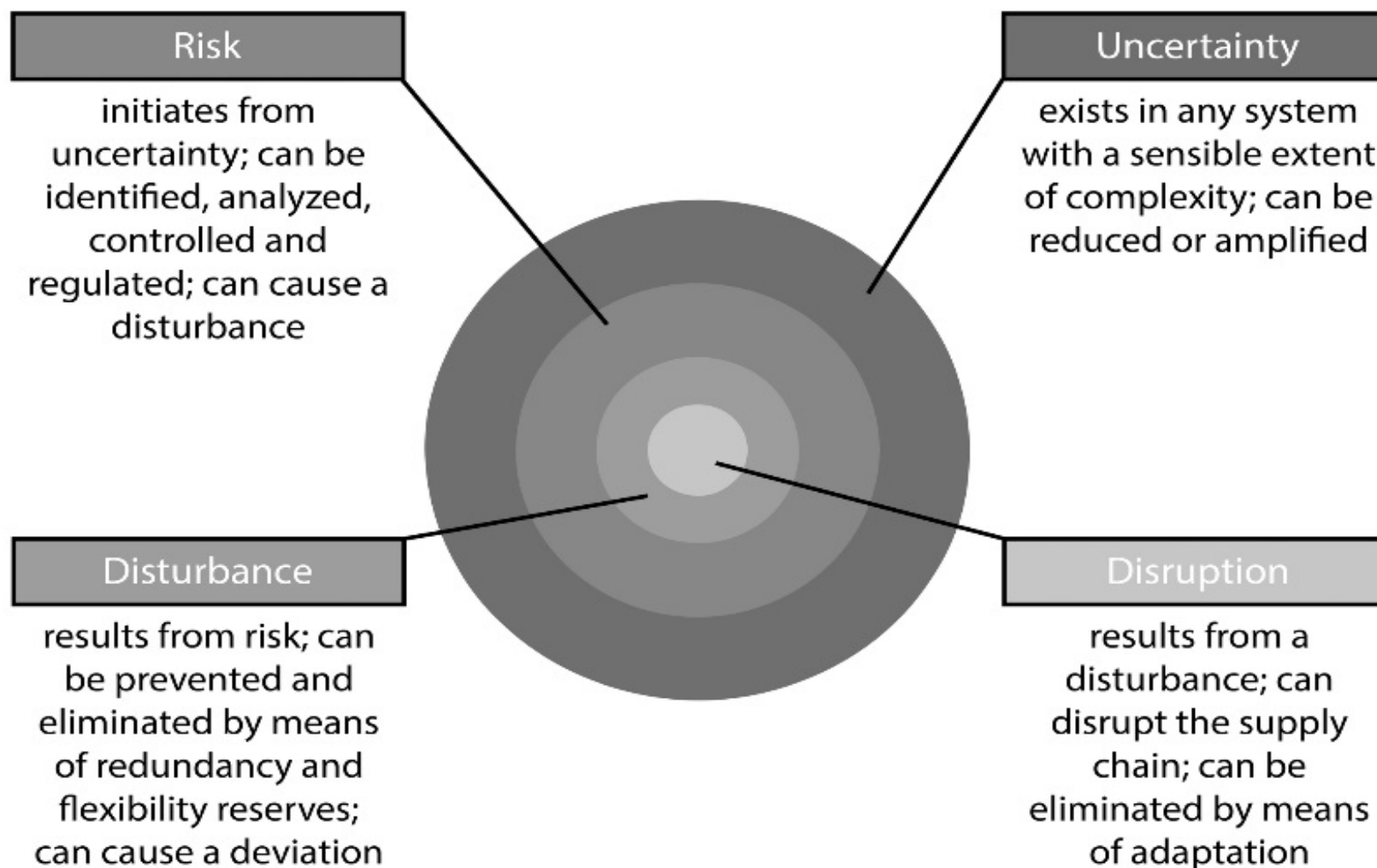


Learning objectives

- Understand RISK.
- Types of risks.
- Identify and quantify impacts.
- Methodology to manage risks and impacts.
- Risks and sustainability

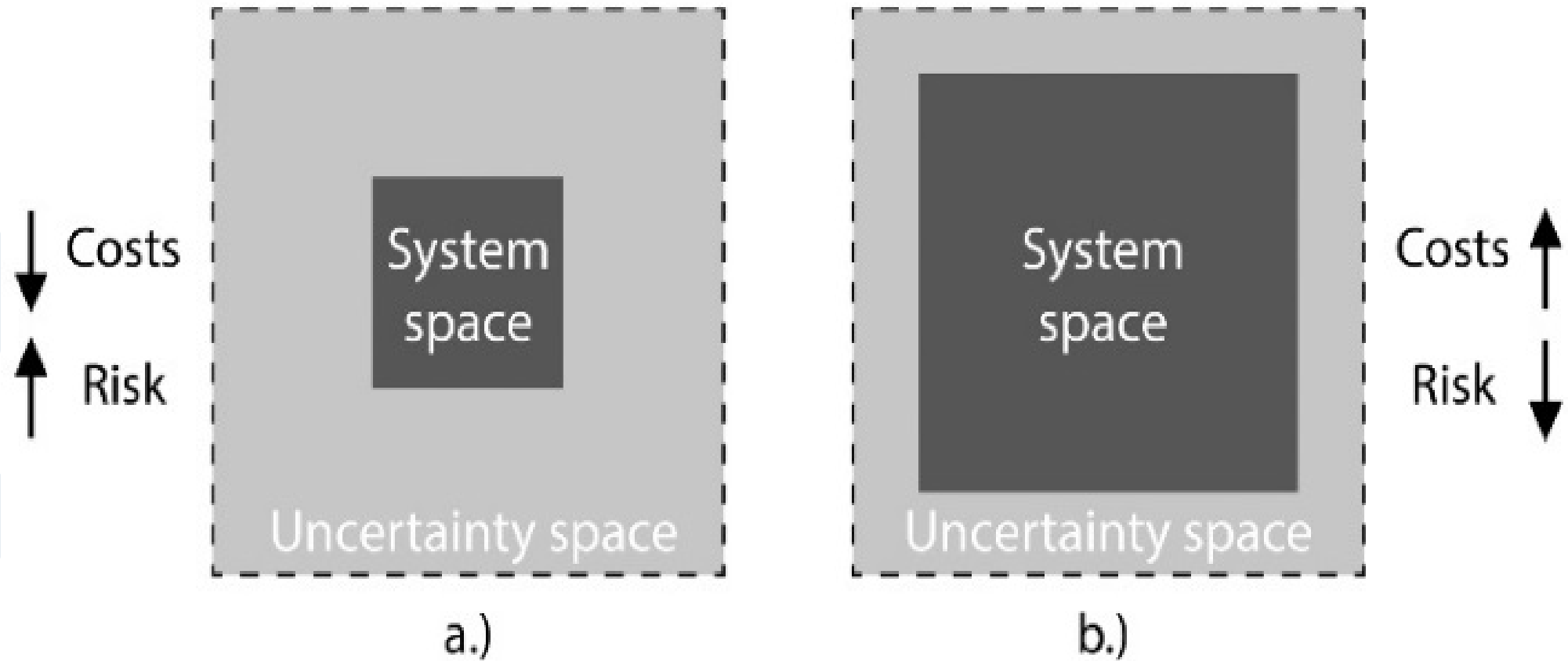


Risks and Uncertainty





Balancing risks and costs





Risk and Supply Chain Management

- **Risk** is a measure of the set of possible (negative) outcomes of a single rational decision and their probabilistic values
uncertainty → probability or occurrence
- Two main Risk categories:
 - Recurring (operational) risks.
 - Disruptive risks.
- Supply chain risk management has become one of the most important topics in practice over the past two decades, especially with respect to operational and disruption risks.
- In supply chain design and planning, we must take uncertainty and risk into account when developing problem approaches and decision-oriented solutions.

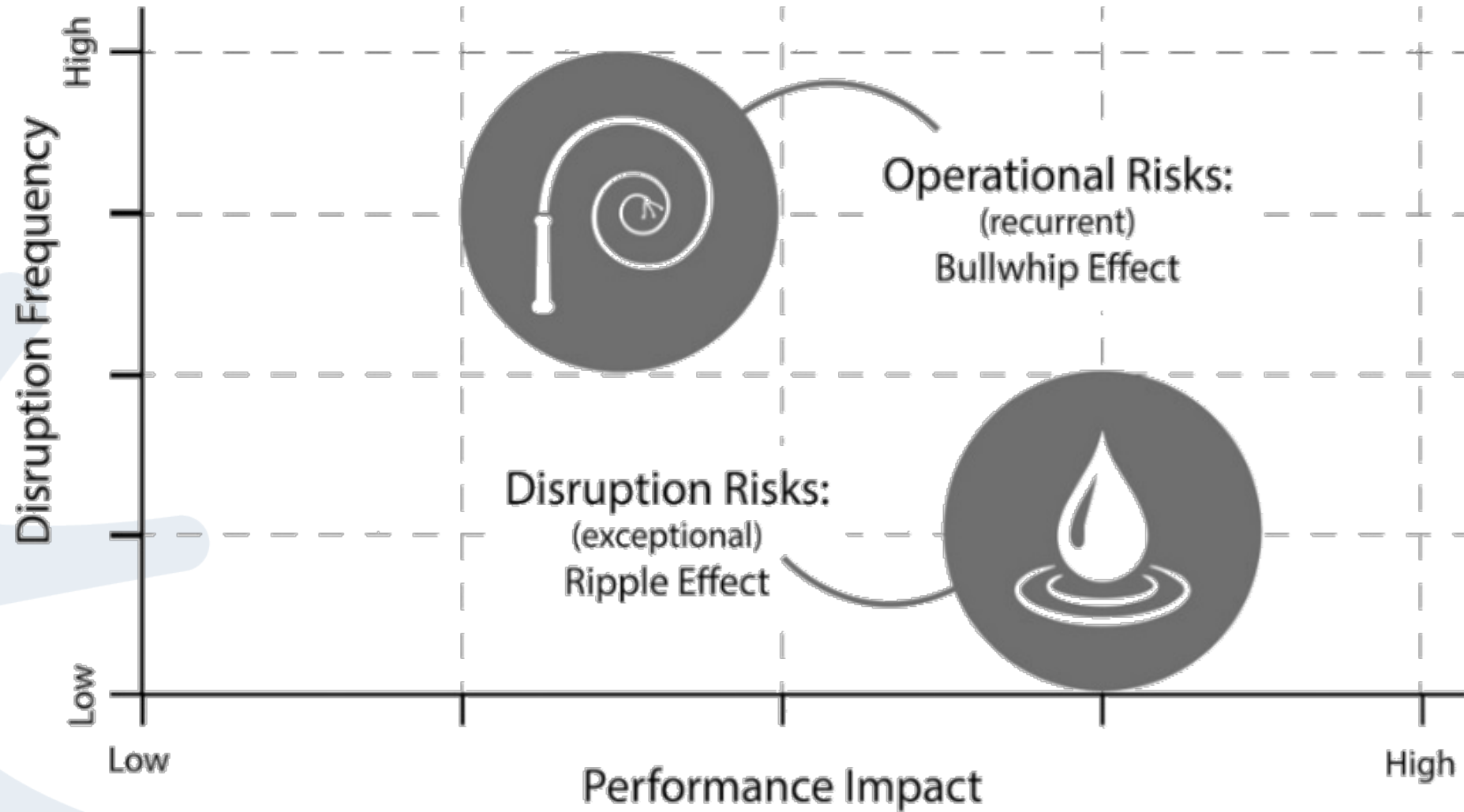


Risk and impacts

- Risks are transformed into impacts that will affect performance based on their frequency and impact:
 - High-frequency, low-impact interruptions are considered to be due to the BULLWHIP effect and refer to fluctuations in demand and delivery time.
 - The BULLWHIP effect considers weekly/daily demand and lead time fluctuations as the main drivers of supply chain changes that take place at the parametric level and can be eliminated in a short-term perspective.
 - The RIPPLE effect in low-frequency, high-impact outages has an important effect because of its cascading impact.

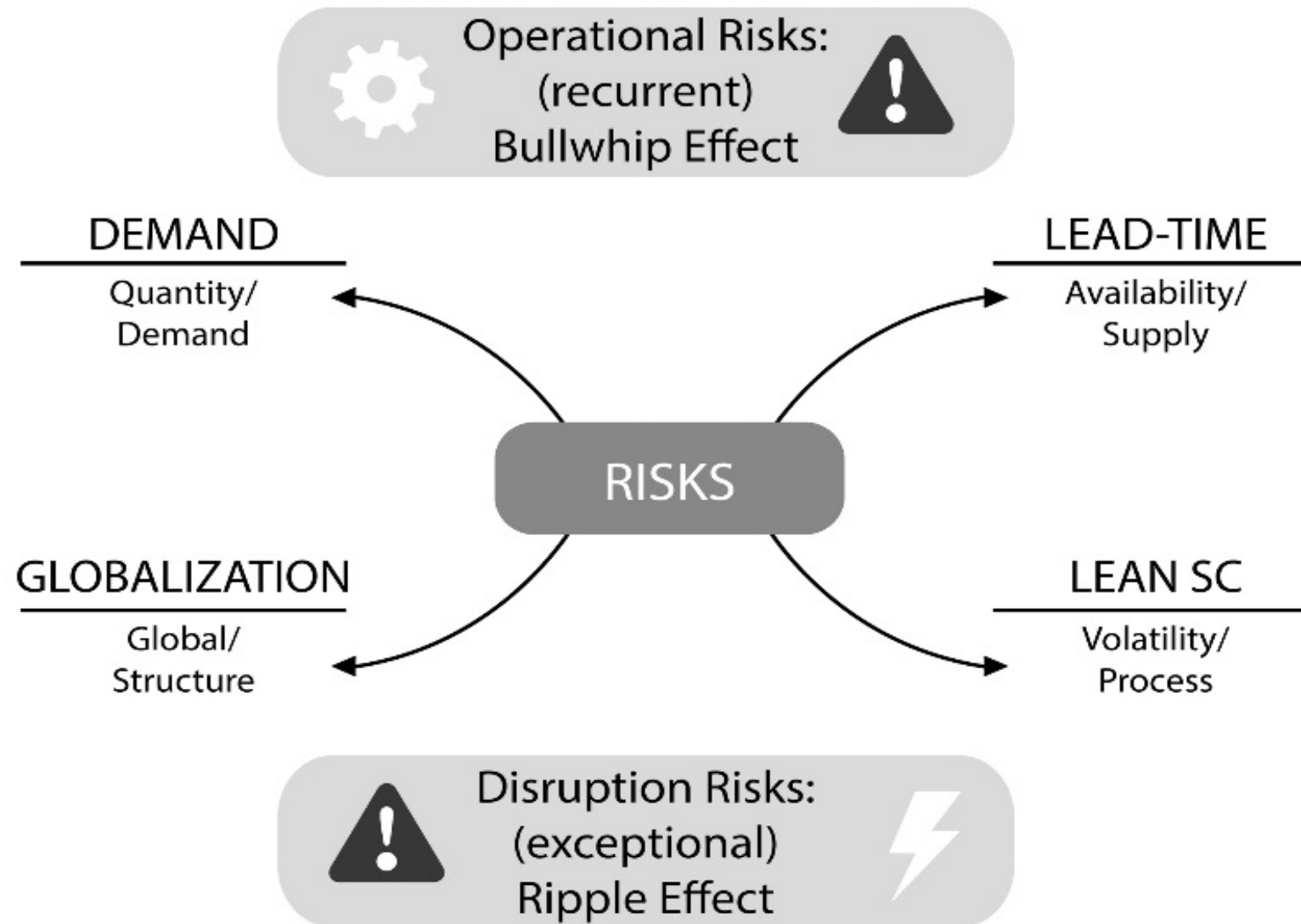


Impact and Frequency





Risks Taxonomy in the Supply Chain





Probability – Impact Matrix

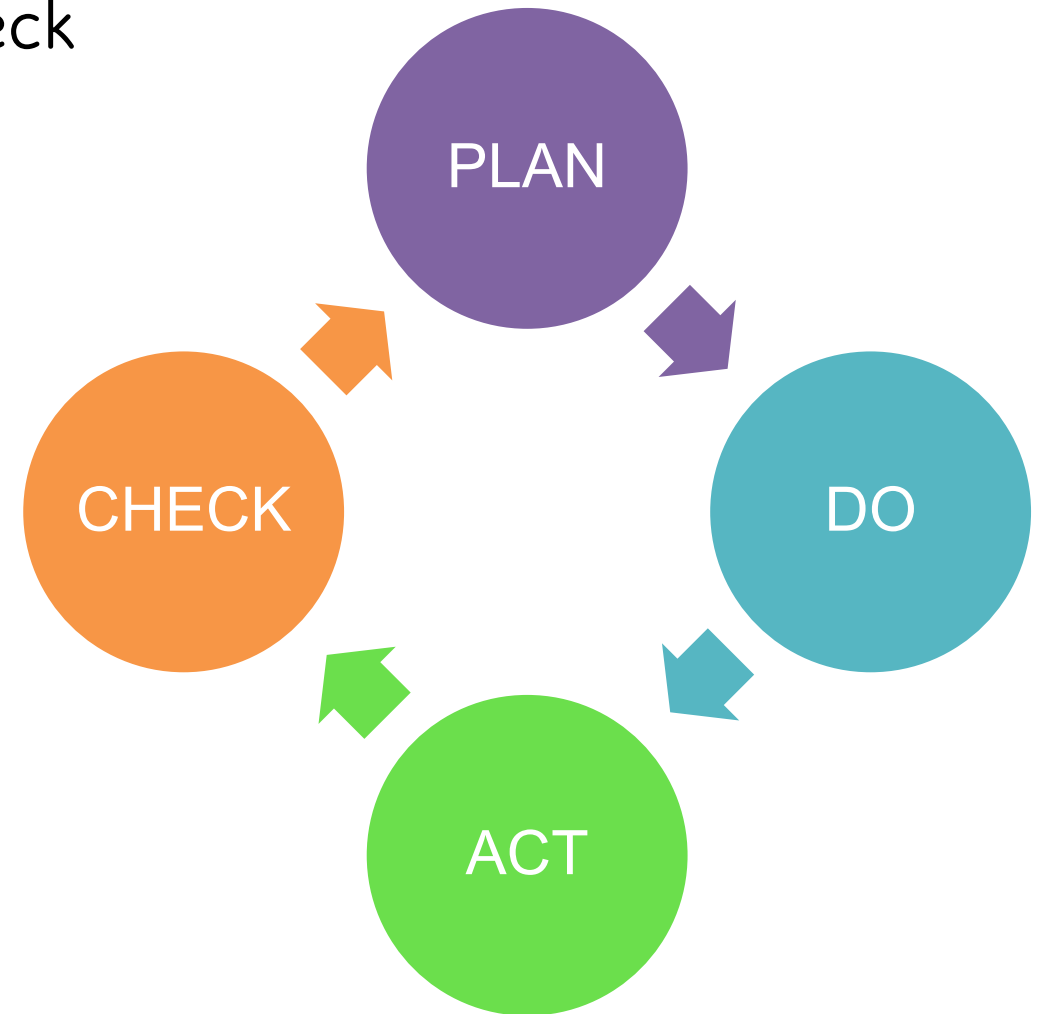
Impact	Critical					★		★
	Very High	★	★		★			★
	High				★	★		
	Mid		★			★	★	
	Very Low			★		★		
	Low			★				
	None	★		★			★	
		Impossible	Very Low	Low	Mid	High	Very High	Critical or Certain

★ Enumerated risk



Risk assessment basic methodology

- Deming cycle: Plan – Do – Act – Check
- Plan:
 - Enumerate risks
 - Quantify by simulation
 - Create plans
- Do:
 - Simulate the plans
 - Implement the plans
- Act:
 - React to impacts
- Check





Digital Twins and Simulators

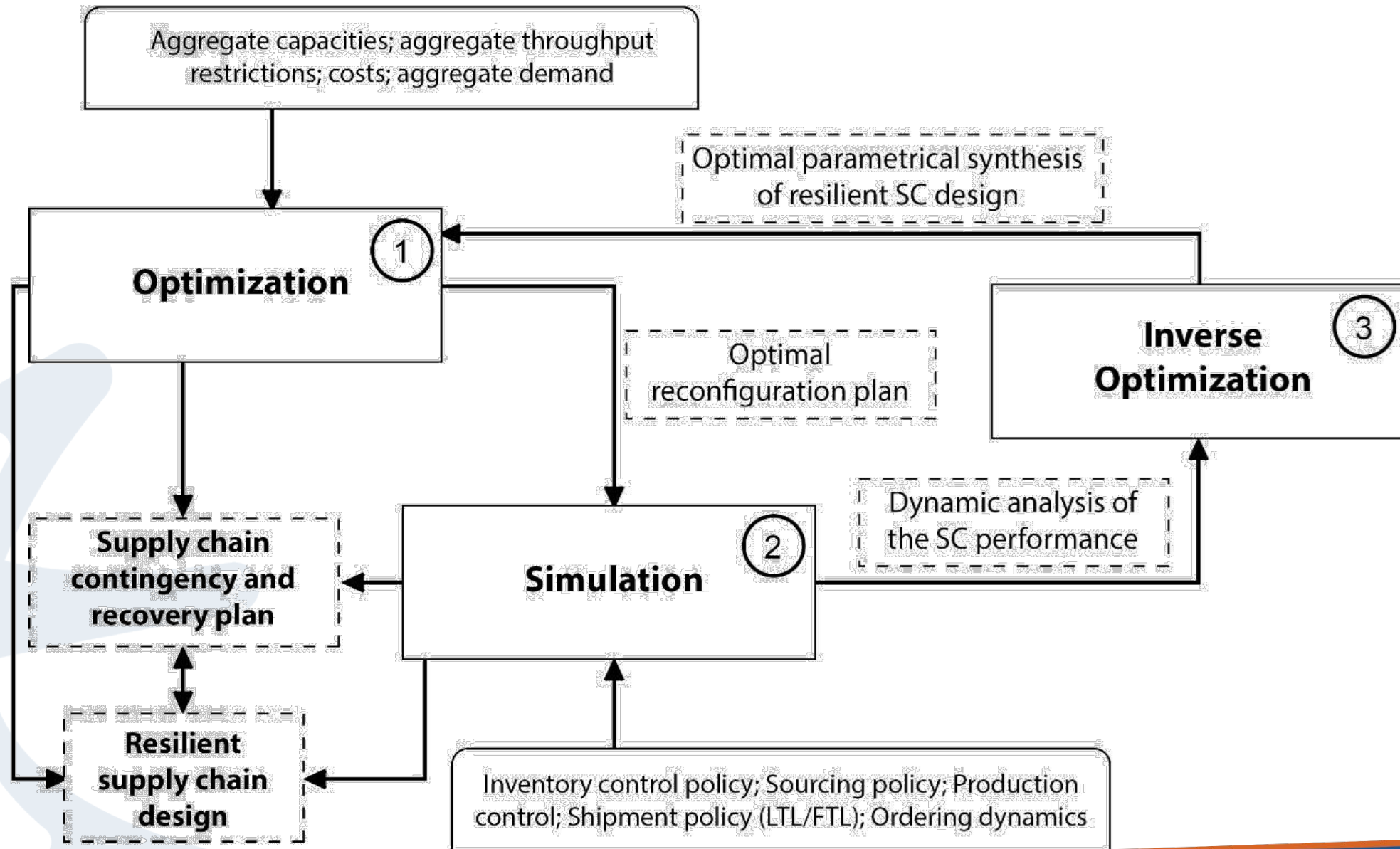
- It is feasible to understand and list the possible risk causes
- Probability can be derived from historical data, experience, or expert knowledge.
- But... how to quantify the possible impacts?
 - Consider that RISKS can occur in a combined way.
 - Consider that many RISK initiate a RISK CHAIN.



- Simulation based on Digital Twins allows us to calculate precisely the real impacts that certain event can bring into the supply chain.



Risk management through simulation





The RIPPLE effect

- The RIPPLE effect appears when changes in the supply chain occur at the structural level as a result of natural and man-made disasters.
- Structural changes propagate through the CoS and recovery can take medium to long term time periods with a significant impact on production performance, such as annual revenues.
- Usually causes low-frequency, high-impact disruption, usually an exceptional risk



The BULLWHIP effect

- Opposite of the RIPPLE effect.
- It considers low frequency-high impact risks, which are operational and recurrent.
- It is the result of the propagation of an initial disturbance to other SC stages in the supply, production and distribution networks.



Ripple vs. Bullwhip

	RIPPLE Effect	BULLWHIP Effect
Risks	Disruptions (e.g. Fire)	Operational (e.g. demand fluctuation)
Affected areas	Structure and critical parameters (unavailability, lost sales, etc).	Operations: lead-time, inventory, profitability, etc.
Recovery	Medium-long term. Considerable effort and investment. Strategy changes.	Short-term coordination to balance demand and delivery.
Impact on efficiency	Impact on annual accounts and profits.	On-time and on-going efficiency, in the form of stock-out, cost overruns, etc.



Spurious or Structural ?

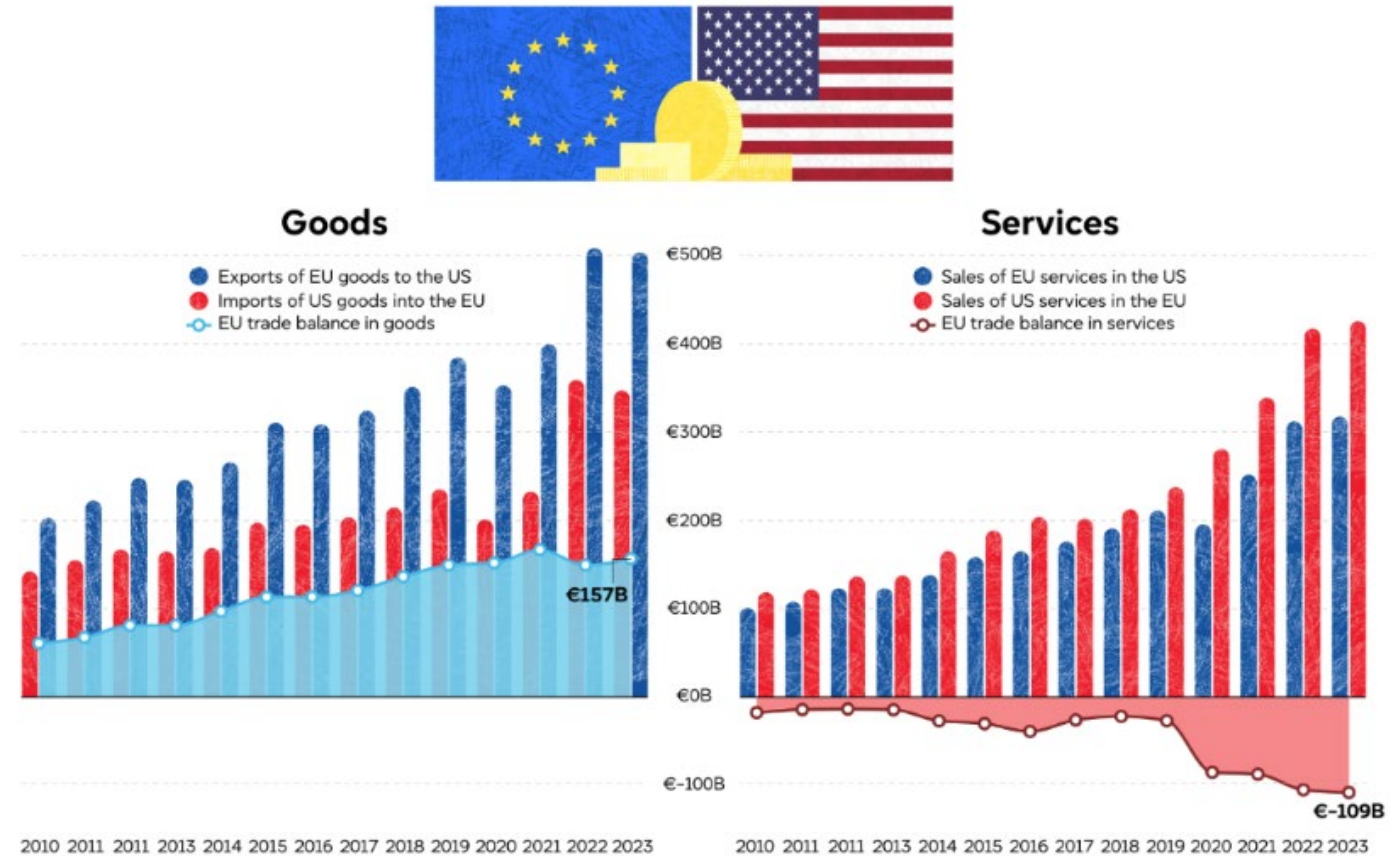
- Not always clear...
- ... But always clear that protection is required.



March/25: Effects of new US tariffs on the EU

<https://www.europarl.europa.eu/topics/en/article/20250210STO26801/eu-us-trade-how-tariffs-could-impact-europe>

Trade in goods and services between the EU and the US 2010-2023, in billions of euros





March/25: Effects of new US tariffs on the EU

In this article

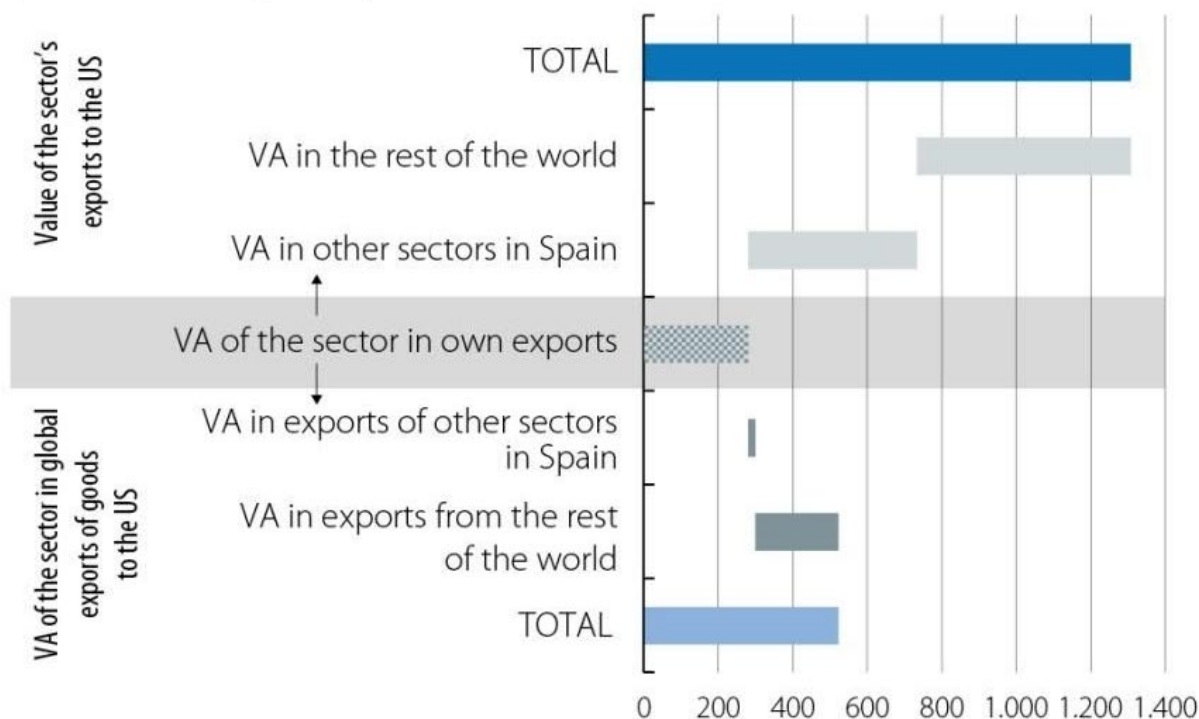
The position in value chains and production specialisation as key factors

Utilities, trade, transport, and professional and auxiliary services, among the hardest hit

Recent trends in US imports suggest a slight increase in exposure in the EU

Value of automotive exports from Spain to the US and value added of the sector in global exports of goods to the US

(USD millions, 2019)



Source: CaixaBank Research, based on data from the OECD.

<https://www.caixabankresearch.com/en/economics-markets/activity-growth/exposure-european-economy-us-tariff-hike-perspective-through>



March/25: Effects of new US tariffs on the EU



Insights

Industries

Services

Events

Careers

About us

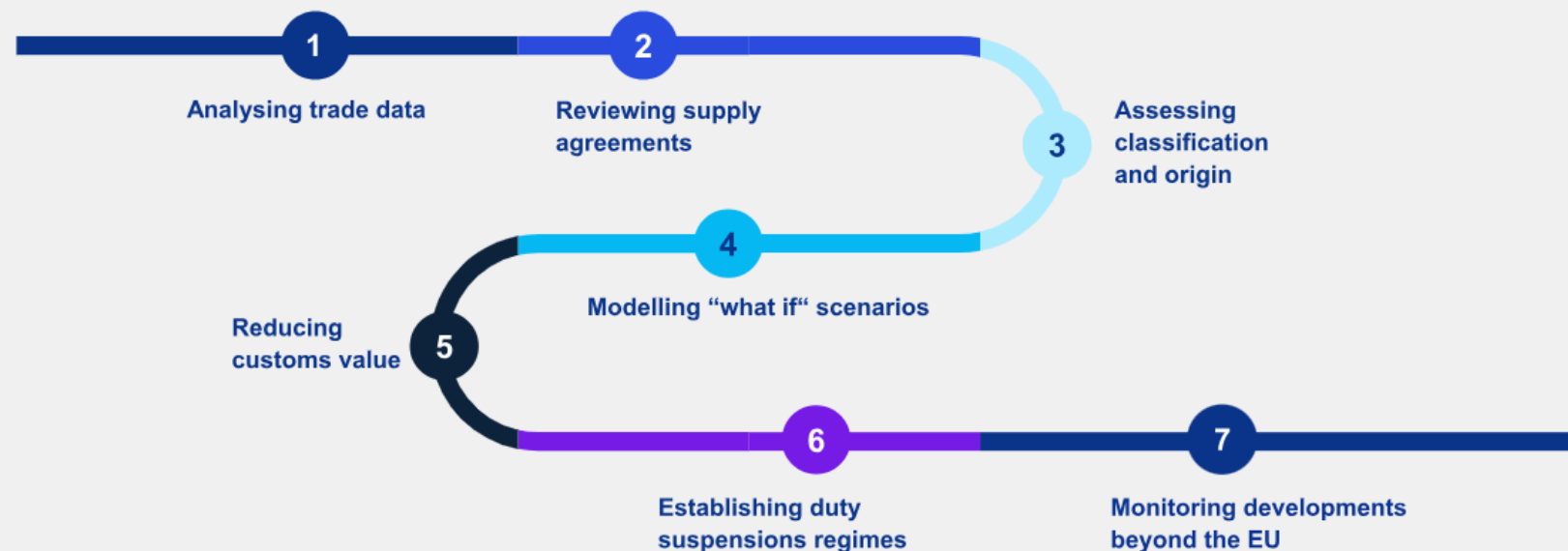
US tariffs: consequences for German...



What should companies do?

As a first step, we recommend that companies gain a complete and detailed understanding of their supply chain. They should be aware of what type of products are sourced from suppliers, which country these products come from and how announced measures may affect purchases and sales.

We have developed a tool that can provide a detailed insight into your flow of goods based on your customs data - [click here for more information](#).



<https://kpmg.com/de/en/home/insights/2025/04/us-cell-consequences-for-german-companies.html>



April/25: Spain-Portugal Electrical Black-Out

BBC

Home News Sport Business Innovation Culture Arts Travel Earth Audio Video Live

Spain and Portugal scramble to restore power as officials chase cause of outage

8 hours ago

Share Save

Gabriela Pomeroy
BBC News



FINANCIAL TIMES

HOME WORLD US COMPANIES TECH MARKETS CLIMATE OPINION LEX WORK & CAREERS LIFE & ARTS HTSI

Spain [+ Add to myFT](#)

Spain declares state of emergency in the wake of huge power outage

Many face night without electricity after train network, traffic lights and mobile connections shut down by mystery blackout

David Sharrock in Madrid, Andy Bounds in Brussels, Ian Johnston in Paris and Rachel Millard in London

Published YESTERDAY | Updated 23:37

972



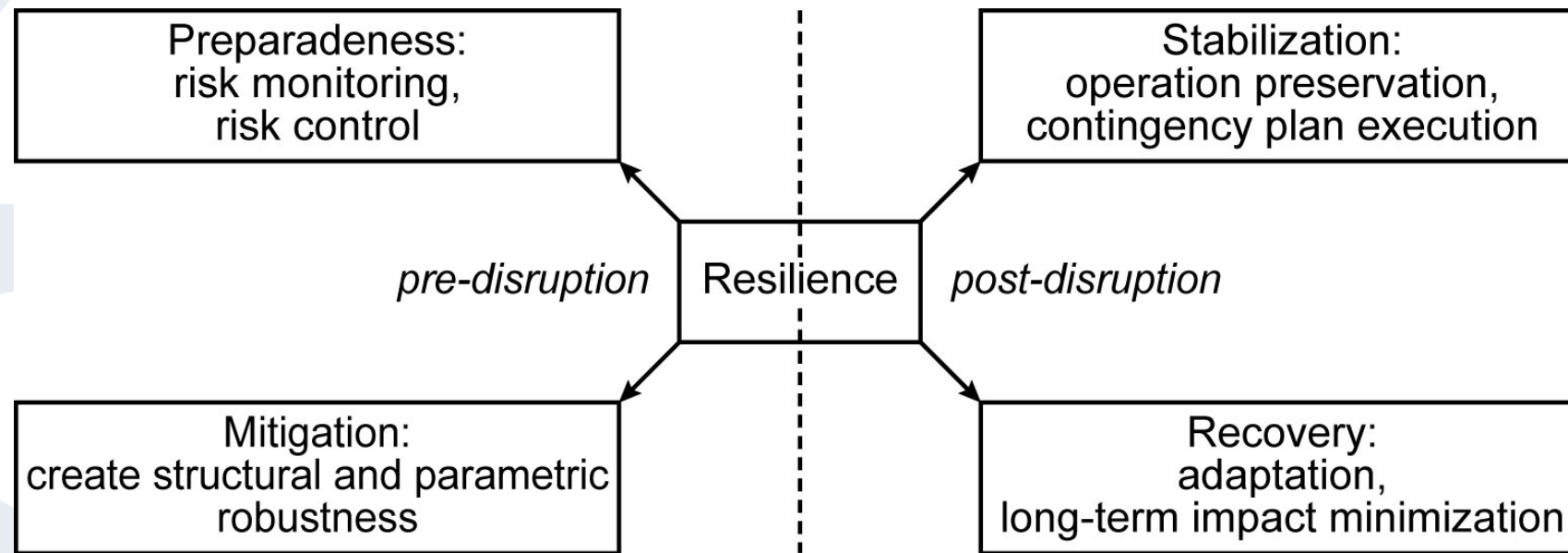


Resilient Supply Chains



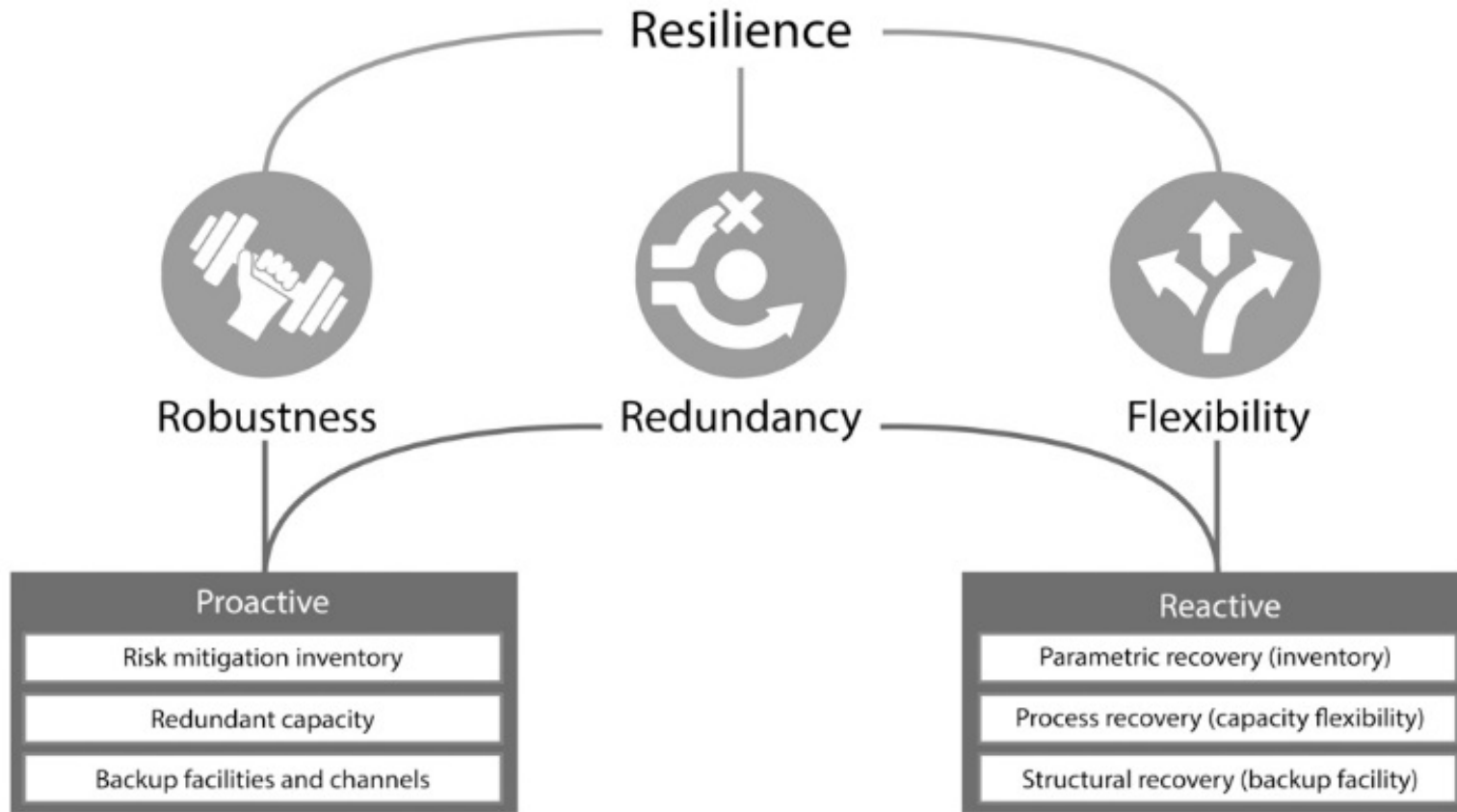
Supply chain resilience

Resilience is the ability to maintain, execute, and recover (adapt) planned performance along with the achievement of planned (or adapted, but still acceptable) performance. Building a resilient supply chain is based on risk mitigation, disruption preparedness, stabilization, and recovery.





Resilience and robustness



- **Robustness** is based on redundancy to guarantee a certain level of performance across a wide range of uncertainty.
- **Resilience** is based on redundancy and flexibility.
- **Flexibility** is a system's ability to adapt to situational processes and structural changes as a reaction to internal and external disturbances.